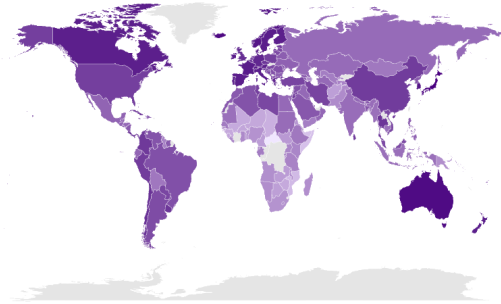


Automated Data Analysis & Reporting

Relationship between Life Expectancy and Child Mortality

Amina Ben Nejma 28.02.2026

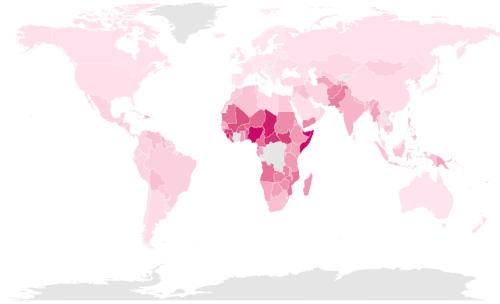
Life Expectancy at Birth - 2020



Life Expectancy
(years)

60 70 80

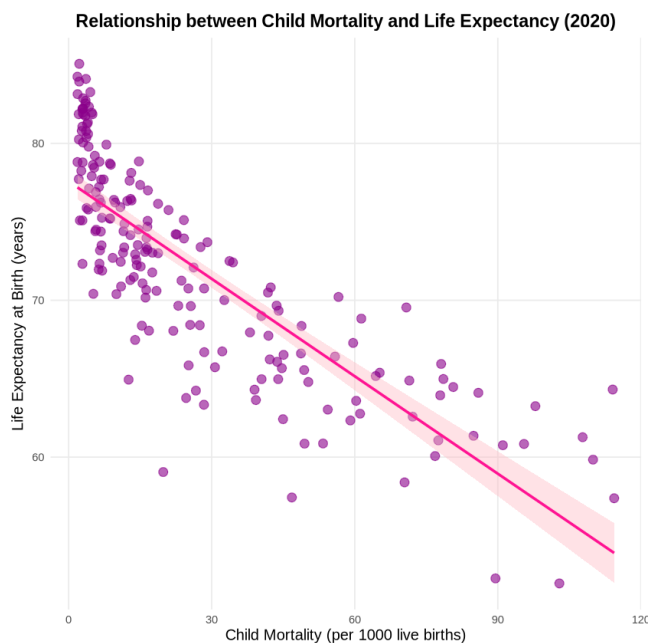
Child Mortality (0-5 years) - 2020



Child Mortality
(per 1000 births)

30 60 90

The maps clearly show geographic patterns - higher life expectancy (darker violet) in developed regions like North America, Europe, and Australia, while lower life expectancy appears in parts of Africa. The child mortality map shows the inverse pattern, with darker pink (higher mortality) concentrated in sub-Saharan Africa.



The analysis reveals a **strong negative relationship** between child mortality and life expectancy:

- Correlation coefficient: -0.82 - This indicates a strong inverse relationship
- R-squared: 0.67 - Child mortality explains about 67% of the variation in life expectancy
- Regression equation: Life Expectancy = 77.58 - 0.21 × Child Mortality

Key findings:

- For every additional child death per 1000 births, life expectancy decreases by approximately 0.21 years
- Countries with low child mortality (under 10 per 1000) tend to have life expectancies above 70 years
- Countries with high child mortality (above 50 per 1000) typically have life expectancies below 65 years
- The relationship is statistically significant ($p < 0.001$)

This strong negative correlation makes intuitive sense: both indicators reflect overall healthcare quality, sanitation, nutrition, and socioeconomic development. Countries that successfully reduce child mortality through better healthcare and living conditions also tend to have higher overall life expectancy.

Reflection on the interaction with the AI tools:

I was surprised on how easy it was to work with Julius. First of all, the processing of the prompt is really. Second, the outcome was exactly how I wanted it, expect minor mistakes, which were easily fixed by being more specific. I didn't think it would work as fast as it did, as for me ChatGPT, which I usually use, takes a pretty long time to work on my prompts. Contrary to ChatGPT it also didn't crash my Google Chrome which I really appreciate. What I did learn is to use the right AI Tool for my work and don't just use ChatGPT for everything, as I now experienced that other tools might be faster and easier.